The equations of general relativity, which underlie the standard big bang model, show that if the matter/energy density early on was *exactly* equal to the critical density, then it would stay equal to the critical density as space expanded.[17] But if the matter/energy density was even slightly more or slightly less than the critical density, subsequent expansion would drive it enormously far from the critical density. Just to get a feel for the numbers, if at one second ATB, the universe was just shy of criticality, having 99.99 percent of the critical density, calculations show that by today its density would have been driven all the way down to .00000000001 of the critical density. It's kind of like the situation faced by a mountain climber who is walking across a razor-thin ledge with a steep drop off on either side. If her step is right on the mark, she'll make it across. But even a tiny misstep that's just a little too far left or right will be amplified into a significantly different outcome. (And, at the risk of having one too many analogies, this feature of the standard big bang model also reminds me of the shower years ago in my college dorm: if you managed to set the knob perfectly, you could get a comfortable water temperature. But if you were off by the slightest bit, one way or the other, the water would be either scalding or freezing. Some students just stopped showering altogether.)

For decades, physicists have been attempting to measure the matter/energy density in the universe. By the 1980s, although the measurements were far from complete, one thing was certain: the matter/energy density of the universe is not thousands and thousands of times smaller or larger than the critical density; equivalently, space is not substantially curved, either positively or negatively. This realization cast an awkward light on the standard big bang model. It implied that for the standard big bang to be consistent with observations, some mechanism—one that nobody could explain or identify—must have tuned the matter/energy density of the early universe *extraordinarily* close to the critical density. For example, calculations showed that at one second ATB, the matter/energy density of the universe needed to have been within a *millionth of a millionth of a percent* of the critical density; if the matter/energy density deviated from the critical value by any more than this minuscule amount, the standard big bang model predicts a matter/energy density today that is *vastly* different from what we observe. According to the standard big bang model, then, the early universe, much like the mountain climber,